made of select polymers and nanowires. This could potentially lead to improvements in smart clothing, electronic skin and other applications that require pliable circuitry.

Lead author Shenqiang Ren, professor in Department of Mechanical and Aerospace Engineering at the university, says, "Traditional electronics, like PCBs in electronic devices, are rigid. That is not a good match for the human body, which is full of bends and curves, especially when we are moving.

"We examined the design principles behind Kirigami, which is an efficient and beautiful art form, and applied those to our work to develop a much stronger and stretchable conductor of power."

The study, which includes computational modelling contributions from Temple University (Pennsylvania, USA) researchers, employs nanoconfinement engineering and strain engineering, which is a strategy in semiconductor manufacturing used to boost device performance.

A wearable mouse from Vicara

Vicara, a startup from India, released a gesture-driven wearable named Kai. This wearable allows you to use your fingers to execute commands on devices like computers and drones. Whether it is designing, video editing or gaming, Kai adds a whole new dimension to your control, making your interactions easy, intuitive and fun. The wearable tracks your fingers and wrist movements effortlessly. It is designed to adapt to your gestures over time, making it more intuitive and easy to use over time. Using Kai control centre, you can pair your favourite gestures to do the things you want in your favourite applications. This makes your workflow more efficient, intuitive and productive.

Kai's slim, band-like design makes it easier to fit into your pocket, pouch or backpack (Credit: https://getkai.co/)

Google's AR-powered microscope to help in real-time cancer detection

Machine learning has the potential to fuel major technological developments in a number of fields. Researchers at Google have developed an augmented reality (AR)-powered microscope prototype that utilises AR to help physicians diagnose cancer. This prototype can analyse biological tissue samples to automatically detect anomalies caused by cancerous cells in the body.

As per the press release from Google, the scientists modified a standard optical microscope to have built-in real-time AI capabilities. This required integrating an AR display into the optics, training a deep neural network for cancer detection and running the AI algorithms connected to the microscope with low latency. As sample applications, they trained AI algorithms to detect metastatic breast cancer in sentinel lymph nodes and prostate cancer in prostatectomy specimens.

Integration of projection optics into an existing microscope provided a parallax-free, high-resolution display of information derived by machine learning overlaying the sample. The AI algorithm ran in real-time using live image feed from the camera, and generated high accuracy cancer detection results. Pathologists testing the device reported a seamless experience that provided immediately useful information.

Tech that allows you to feel VR

Virtual reality (VR) has seen tremendous growth within the tech industry. Augmented reality and VR eyewear are projected to grow by 25 per cent to sell 4.9 million units in 2018, according to Consumer Technology Association (CTA).

HaptX Gloves feature over 100 points of high-displacement tactile feedback, up to five pounds of resistance per finger and sub-millimetre precision motion tracking (Credit: https://haptx.com)

HaptX, a CTA member, is taking VR to the next level by incorporating a sense of touch. HaptX Gloves bring realistic touch, powerful force feedback and precise motion tracking to VR. Applications for the technology are endless, from training programs to an elevated entertainment experience.

HaptX's patented microfluidic technology lets you feel the shape, movement, texture and temperature of digital objects. It can be integrated into any garment—from gloves to a full-body suit—to deliver unprecedented realism in virtual experiences.

Get e-mail on your skin, says Facebook

Scientists at Facebook have created a wearable device that allows people to feel words on their arms. The prototype of a cast-like wearable device is filled with actuators that, when triggered, cause vibrations on the arm in patterns that match certain sounds.

Researchers were able to teach people to feel four different phonemes—individual sounds that make up words in a language—in three minutes. Over a period of more than 90 minutes of training, participants were able to learn to recognise 100 words, as per Ali Israr, technical lead for the project.